

Design and Techno-Economic Analysis of In-House Quasi-Direct-Drive Actuators for Humanoid Robots in India

JX1 Humanoid Project

Abstract—Joint actuators account for 77% of the parts cost of the JX1 humanoid and are today imported from China. This paper asks whether quasi-direct-drive (QDD) actuators of the RobStride class, and the larger units needed by full-size humanoids, can be designed and built in India with minimal tooling, and at what cost. We (i) derive joint torque requirements from a rigid-body analysis of JX1, the published limits of full-size humanoids and human biomechanics, arriving at three classes of 40, 120 and 360 N·m peak torque; (ii) size an outer-rotor surface-permanent-magnet motor with tooth-concentrated windings and a single-stage planetary reducer for each class with an analytic electromagnetic, thermal and gear-stress model, which reproduces the RobStride RS04 datasheet within 3–20%; (iii) describe a winding, casting and assembly process that relies on Indian job shops rather than a factory; and (iv) build a cost model at prototype, batch and 1,000-unit-per-year scale. The 120 N·m class reaches 125 N·m peak, 46 N·m continuous and 23.1 rad/s at 48 V with an estimated mass of 1.91 kg. Built in batches of 25 it costs INR 30,292 per unit against INR 24,091 for an imported RS04, and an Indian line breaks even with the import only at about 875 units per year. In-house production is therefore justified by capability, the 360 N·m class and supply security, not by unit cost at small volume.

Index Terms—Humanoid robot, quasi-direct-drive actuator, permanent-magnet motor, concentrated winding, planetary gear, techno-economic analysis, India.

I. INTRODUCTION

LEGGED and humanoid robots are built around their joint actuators. Since the MIT Cheetah [1], [2], the dominant design for dynamic legs has been the *quasi-direct-drive* (QDD) actuator: a large-diameter permanent-magnet motor with a low-ratio (6–10:1) planetary reducer, which keeps the joint back-drivable, tolerant of impacts and torque-controllable through motor current. Chinese manufacturers such as RobStride, Damiao and Unitree now sell such actuators at about US\$1–2 per newton-metre of peak torque [10], [29], which has made low-

cost humanoids possible, including the 1.2 m, 27.5 kg JX1 developed in this project.

For an Indian builder, however, the actuators are imported (77% of the JX1 bill of materials), subject to duty, lead time and export-control risk, and no Indian manufacturer offers comparable units at volume [30]. This motivates three questions:

- 1) How much torque and speed do humanoid joints need, from JX1 up to 70 kg full-size robots?
- 2) Can a QDD actuator of that class be designed from first principles and built with job-shop processes and hand winding?
- 3) What does it cost compared with importing, and compared with what Chinese manufacturers spend?

The contributions of this paper are: an open, reproducible analytic sizing model for outer-rotor QDD actuators validated against a commercial datasheet; a three-class actuator family (JXA-40/120/360) with complete winding data; a manufacturing route, including a casting and moulding analysis, suited to Indian job shops; and a make-versus-buy cost model at three production scales.

II. BACKGROUND AND RELATED WORK

A. Electromagnetic principles

A current-carrying conductor of length ℓ in a flux density B experiences the Lorentz force $F = BI\ell$, and a changing flux linkage induces the back-EMF $e = -d\lambda/dt$ (Faraday). Integrating the tangential force over the air gap gives the torque of any radial-flux machine in terms of the air-gap shear stress σ , gap diameter D and stack length L [5]:

$$T = \frac{\pi}{2} \sigma D^2 L. \quad (1)$$

Air-cooled servo machines sustain $\sigma \approx 5\text{--}15$ kPa continuously and 30–60 kPa for seconds. Because torque grows with D^2 but only linearly with L , robot actuators are flat “pancake” machines, usually with an outer rotor that maximises D for a given envelope [6]. Field-oriented control (FOC) keeps the stator field in quadrature with the rotor magnets, so that torque is proportional to the q -axis current.

B. Quasi-direct-drive actuators

Wensing *et al.* [1] and Katz [2] showed that a 6:1 single-stage planetary with a large-gap motor gives the impact

Manuscript prepared October 2026. This work is part of the open-source `thenar-jx1` humanoid repository (<https://github.com/nickthelegend/thenar-jx1>); all design scripts, cost models and research logs cited here are in `actuators/inhouse/` and `research/raw/`.

Labels: values marked *calculated* come from the analytic model of Section IV; *estimated* costs are derived from cited price listings (24 Sep–6 Oct 2026); nothing in this paper has yet been built or measured.

tolerance and force transparency needed for dynamic locomotion; the MIT Humanoid [3] scaled the concept to 33.6 and 68 N·m modules. A gear ratio G multiplies torque by G but reflected rotor inertia by G^2 , which is why legged robots favour 6–10:1. Open-source low-cost designs include Berkeley Humanoid Lite [4] with 3D-printed cycloidal reducers. Table I summarises representative commercial and research actuators; the full survey of about 110 units is in the project logs [26].

No teardown of a RobStride actuator has been published. The vendor's own site data give 42 poles, a 9:1 machined-steel planetary and the electrical constants used for validation in Section V-B; every 42-pole motor of this size with a published slot count (CubeMars AK10-9, Unitree Go2, T-Motor U10) is 36N42P [12], [13].

III. TORQUE AND SPEED REQUIREMENTS

A. JX1 (1.2 m, 27.5 kg)

A rigid-body inverse-dynamics model of JX1 walking at 0.30–0.79 m/s, turning, squatting and standing on one leg, with peak requirement $\max(1.5 \tau_{\text{dyn}}, 1.25 \tau_{\text{static}})$, gives the joint requirements of Table II (calculations/ in the repository).

B. Full-size humanoids and human data

Table III lists published limits of 50–70 kg humanoids from official model files and SDKs [14]–[16]; peak knee torque normalised by robot mass ranges from 2.1 to 7.7 N·m/kg. Human knee moments reach 0.5–1.3 N·m/kg in walking, up to 2.0 in rising from a chair, 3.1 in running at 3.5 m/s and 3.6 in sprinting [19]–[21]. A 70 kg robot that should run or rise from the floor therefore needs 200–360 N·m at the knee.

C. Actuator classes

Three classes cover JX1 completely and every leg joint of the largest humanoids: **JXA-40** (40 N·m, ≥ 40 rad/s; RS06 class), **JXA-120** (120 N·m, ≥ 20 rad/s; RS04 class) and **JXA-360** (360 N·m, ≥ 15 rad/s; Unitree H1 knee class, for which no RobStride product exists).

IV. DESIGN MODEL

Each actuator is an outer-rotor surface-PM motor with double-layer tooth-concentrated windings, a single-stage planetary (sun input, ring fixed, carrier output), dual magnetic encoders and a 48 V FOC driver on the rear cover (Fig. 2). The model is implemented in `qdd_design.py`.

A. Magnetic circuit and torque constant

With remanence B_r (N42SH, 80 °C), recoil permeability μ_r , magnet thickness h_m , air gap g , Carter coefficient k_C and leakage factor $k_l = 0.95$, the air-gap flux density and its fundamental are

$$B_g = \frac{k_l B_r}{1 + \mu_r k_C g / h_m}, \quad B_1 = \frac{4}{\pi} B_g \sin\left(\frac{\alpha_p \pi}{2}\right), \quad (2)$$

with pole-arc ratio $\alpha_p = 0.85$. Teeth and yokes are sized for 1.75 T and 1.55 T in M270-35A steel. For p pole pairs, N_{ph} series turns per phase and winding factor k_w , the phase flux linkage and torque are

$$\lambda = \frac{k_w N_{\text{ph}} B_1 D L}{p}, \quad T = \frac{3}{2} p \lambda i_q = K_t i_q. \quad (3)$$

B. Winding factor and layout

For Q slots and $2p$ poles, adjacent tooth coils are displaced by $\theta_s = 2\pi p/Q$ electrical. The pitch factor is $k_p = |\sin(p\pi/Q)|$; the distribution factor k_d and the coil-to-phase assignment follow from the star of slots [7], [8]: each coil phasor is assigned to the 60° phase belt that contains it, and $k_w = k_p k_d$. The resulting layouts (Fig. 3) give $k_w = 0.933$ (24N28P), 0.945 (36N40P) and 0.949 (48N44P), all with an even number of repeating units (no unbalanced magnetic pull).

C. Turns, wire and resistance

The number of turns is chosen from the speed target: $\lambda_{\text{target}} = V_{\text{max}}/(1.12 p G \omega_0)$, with $V_{\text{max}} = 0.95 V_{\text{dc}}/\sqrt{3}$ for space-vector modulation and 12% headroom. The conductor is then made as large as a bare-copper slot fill of 0.42 allows (realistic for careful hand or needle winding). With N_c turns per tooth, a parallel paths, conductor area A_c and mean turn length ℓ_t ,

$$R_{\text{ph}} = \frac{\rho_{\text{Cu}} N_c Q \ell_t}{3a^2 A_c}, \quad K_m = \frac{K_{t,\text{rms}}}{\sqrt{3} R_{\text{ph}}}. \quad (4)$$

The motor constant K_m (torque per square root of copper loss) is independent of the number of turns and is the figure of merit for thermally limited joints.

D. Inductance, torque-speed and thermal limits

The phase inductance is estimated as the sum of the tooth-coil air-gap term, slot leakage and end-winding leakage, $L_{\text{ph}} \approx \frac{Q}{3a^2} \mu_0 N_c^2 \tau_s L/g_e + L_{\text{slot}} + L_{\text{end}}$. With $i_d = 0$ the steady-state voltage limit

$$(R i_q + \omega_e \lambda)^2 + (\omega_e L_{\text{ph}} i_q)^2 \leq V_{\text{max}}^2 \quad (5)$$

together with the current limit gives the torque-speed envelope (Fig. 1). Saturation is represented by an 11% roll-off of K_t between $I_{\text{pk}}/3$ and I_{pk} , calibrated on RS04 (120 N·m at 90 A versus 40 N·m at 27 A). Continuous torque follows from a lumped winding-to-ambient resistance R_{th} (2.00 / 1.05 / 0.62 K/W for a unit bolted to an aluminium limb) and a 120 °C winding limit:

$$3I_{\text{rms}}^2 R_{\text{ph}} + P_{\text{Fe}} = (T_{\text{lim}} - T_{\text{amb}})/R_{\text{th}}. \quad (6)$$

E. Planetary stage

For a planetary with fixed ring, $G = 1 + Z_r/Z_s$, with $Z_r = Z_s + 2Z_p$ and $(Z_s + Z_r)/N_p$ an integer. A 12/42/96 set with three planets gives 9:1 (12/48/108 gives 10:1). The

TABLE I
REPRESENTATIVE QDD AND HUMANOID ACTUATORS (DATASHEET VALUES UNLESS NOTED)

Actuator	Peak [N·m]	Rated [N·m]	No-load [rad/s]	Mass [g]	Reducer / motor	Price
RobStride RS00 / RS02 / RS06 [10]	14 / 17 / 36	5 / 6 / 11	33 / 43 / 50	310 / 405 / 621	10 / 7.75 / 9:1 planetary; 28-pole	US\$125 / 145 / 210
RobStride RS03 / RS04 [10]	60 / 120	20 / 40	20.4 / 20.9	880 / 1,420	9:1 planetary; 42-pole	US\$225 / 255
Damiaio DM-J10010L / J10422P [11]	120 / 400	40 / 100	20.9 / 12.6	1,372 / 2,700	10:1 / 22:1	US\$265 / 480
Unitree Go2 motor [13]	23.7	–	30	530	6.22:1 (9/19/47); outer rotor 36N42P	–
Unitree G1 knee / H1 knee [14]	90–139 / 360	–	~20 / 14	–	G1: 2-stage 22.5:1	–
CubeMars AK80-9 / AK10-9 [12]	18 / 48	9 / 18	– / 10.5	485 / 960	9:1; 36N42P, Ø98 mm	INR 79,739 / 104,809
MIT Humanoid U10 / U12 [3]	33.6 / 68	–	55 / 45	619 / 1,174	6:1 (+ belts)	research
Berkeley Humanoid Lite 6512 [4]	~20–28	–	–	–	15:1 printed cycloid	US\$157–188
Tesla Optimus rotary [17], [18]	20 / 110 / 180	–	–	–	strain-wave + clutch; plus linear 0.5–8 kN	in-house

TABLE II
JX1 JOINT REQUIREMENTS AND THE CLASS THAT COVERS EACH JOINT

Joint	Peak / cont. [N·m]	Today	In-house
Hip pitch	85 / 26	RS04	JXA-120
Knee	71 / 30	RS04	JXA-120
Hip roll	50 / 17	RS03	JXA-120
Hip yaw	37 / 8	RS06*	JXA-40
Ankle motors (×2)	32 / 3.4	RS06	JXA-40
Waist yaw	≈30	RS06	JXA-40
Shoulder, elbow	≤17	RS02 / RS00	buy

*RS06 is 3% below the hip-yaw peak requirement.

TABLE III
JOINT LIMITS OF FULL-SIZE HUMANOID [N·m]

Robot (mass)	Knee	Hip pitch	Hip roll/yaw	Ankle
Unitree H1/H1-2 (47/70 kg)	360	220	220	59 / 2×75
Fourier GR-2/GR-3 (63/70 kg)	366	366	95–140	54–59
UBTech Walker S2 (70 kg)	225	225	225 / 65	2×65
AgiBot A2 Lite (64 kg)	270	–	–	–
Fourier GR-1T2 (≈53 kg)	135	135	84 / 65	42

sun is checked with the Lewis bending and Hertz contact equations [9]:

$$\sigma_F = \frac{F_t K_\gamma K_v}{b m Y}, \quad \sigma_H = Z_E Z_H \sqrt{\frac{F_t(u+1)}{b d_1 u}}, \quad (7)$$

with load-sharing and dynamic factors $K_\gamma = K_v = 1.15$, against case-carburised EN353 ($\sigma_{F,\text{lim}} = 430$ MPa,

$\sigma_{H,\text{lim}} = 1,400$ MPa). The 12-tooth sun needs a +0.3 profile shift to avoid undercut.

V. RESULTS

A. The JXA family

Table IV gives the calculated designs. All three classes meet their peak, continuous and speed targets; peak air-gap shear stresses are 42 / 46 / 45 kPa. Fig. 1 compares the envelopes with commercial units, Fig. 2 shows the JXA-120 section to scale and Fig. 3 the winding layouts.

B. Validation against a commercial datasheet

Table V compares the JXA-120 model with RobStride's published RS04 constants [10]. The calculated torque constant, back-EMF constant, inductance, peak current and no-load speed are within 3–20% of the datasheet; the lower calculated resistance reflects the longer (20 mm) stack and correspondingly more copper. The JXA-40 output torque constant (1.11 N·m/A) likewise matches RS06's 1.10.

C. Discussion of the design

The first-iteration JXA-120 is 34% heavier than RS04, mainly from solid planets, conservative housings and the longer stack; compound planets, lightened carriers, 0.2 mm laminations and die-cast housings are the second-iteration levers. Peak current densities of 60–95 A/mm² are sustainable for only 1.5–4 s (adiabatic copper heating of 40 K), so the controller must enforce an $I^2 t$ limit. The sun gear is the most loaded component (contact stress 1,406 MPa at peak) and must be carburised and ground. The JXA-360 driver must supply 206 A peak at 48 V; a 72 V bus (as on Unitree H1) with 1.5× the turns reduces this by a third.

TABLE IV
CALCULATED JXA DESIGNS (ANALYTIC MODEL; EXPECTED ACCURACY $\pm 15\text{--}30\%$ UNTIL MEASURED)

	JXA-40	JXA-120	JXA-360
Benchmark	RobStride RS06	RobStride RS04	Unitree H1 knee
Slots / poles	24N28P	36N40P	48N44P
Stator OD $\times$ stack [mm]	72 $\times$ 14	100 $\times$ 20	140 $\times$ 28
Rotor OD / stator bore [mm]	82.0 / 43.8	111.1 / 63.7	153.3 / 102.0
Air gap / magnet [mm]	0.5 / 2.5	0.55 / 3.0	0.6 / 3.5
Tooth / slot depth / yoke [mm]	5.3 / 11 / 2.2	5.0 / 15 / 2.2	5.3 / 15 / 2.8
B_g at 80 °C [T]	0.94	0.95	0.96
Winding k_w	0.933	0.945	0.949
Turns/tooth, strands $\times$ wire, paths	7, 2 $\times$ 0.63 mm, 1	19, 1 $\times$ 0.60 mm, 4	4, 3 $\times$ 0.85 mm, 2
Copper fill (bare)	0.41	0.41	0.41
Gear $Z_s/Z_p/Z_r$, module [mm]	12/42/96, 0.7	12/42/96, 1.0	12/48/108, 1.25
Gear ratio G	9	9	10
Peak torque [N·m]	41	125	371
Continuous torque [N·m]	13	46	147
No-load speed at 48 V [rad/s]	47.7	23.1	18.6
Peak phase current [A]	59	86	206
K_t at output [N·m/A _{rms}]	1.11	2.31	2.86
K_e [V _{rms,LL} /krpm]	7.9	16.3	18.2
K_m motor / output [N·m/ $\sqrt{W}$]	0.25 / 2.1	0.62 / 5.3	1.40 / 13.3
R_{ph} (20 °C) / L_{ph}	69 m Ω / 72 μ H	48 m Ω / 85 μ H	12 m Ω / 24 μ H
J peak / cont. [A/mm ²]	95 / 19	76 / 18	61 / 15
Time at peak, adiabatic [s]	1.5	2.3	3.6
Peak air-gap shear [kPa]	42	46	45
Gear SF: bend. static / fatigue / contact	3.4 / 4.6 / 1.89	3.0 / 3.5 / 1.64	2.7 / 2.9 / 1.52
Reflected inertia [kg·cm ²]	192	749	4173
Envelope $\varnothing \times$ length [mm]	90 $\times$ 58	119 $\times$ 67	161 $\times$ 83
Mass [kg] (N·m/kg)	0.88 (47)	1.91 (66)	4.27 (87)

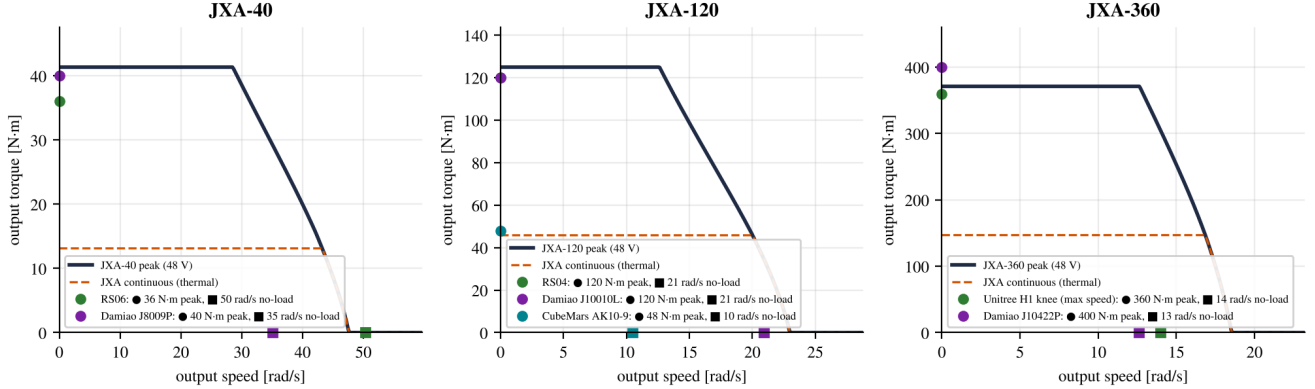


Fig. 1. Calculated torque–speed envelopes at 48 V (no field weakening) and thermal continuous torque, with commercial benchmarks (● peak torque, ■ no-load speed).

TABLE V
MODEL CHECK: JXA-120 VERSUS THE RS04 DATASHEET

Quantity	RS04	JXA-120	Δ
K_t output [N·m/A _{rms}]	2.10	2.31	+10%
K_e [V _{rms,LL} /krpm]	16.9	16.3	−3%
R_{ph} [m Ω]	≈ 80	48	more Cu
L_{ph} [μ H]	≈ 105	85	−19%
Peak current [A]	90	86	−4%
No-load speed [rad/s]	20.9	23.1	+11%
Mass [kg]	1.42	1.91	+34%

D. Prototype shortcut: commercial stator cores

Bare outer-rotor stator cores are sold in China in exactly these sizes [27]. Re-running the model on a 10020 core (100 $\times$ 20 mm, 36N42P, US\$19–31) gives 124/44 N·m at 23.8 rad/s (14 turns, 3 $\times$ 0.4 mm, 3 paths); three stacked 13710 cores (137 $\times$ 30 mm, 36N42P) give 372/147 N·m at 18.0 rad/s (8 turns, 6 $\times$ 0.5 mm, 3 paths). Buying cores removes the lamination tooling step from the first prototypes.

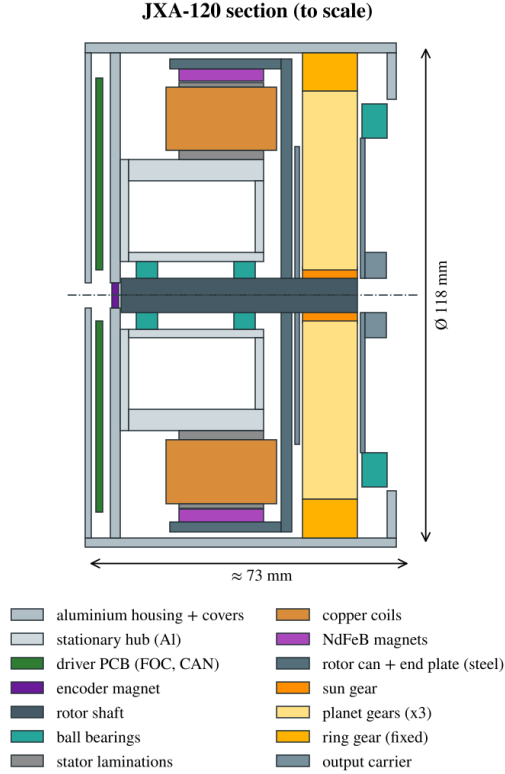


Fig. 2. JXA-120 section drawn to scale from the calculated dimensions: stator on a fixed hub, outer magnet rotor driving the sun gear, ring gear integral with the housing, carrier output on a single large bearing, driver and encoders on the rear cover.

VI. MANUFACTURING ROUTE

A. Winding procedure

Table VI gives the winding data. The procedure, suitable for a technician or a local motor-rewinding shop, is: (1) deburr the lamination stack and insulate it with 0.2–0.3 mm epoxy powder coat or Nomex/DMD liners, verified at 500 V; (2) mark tooth 1 and follow the pattern, winding each phase group with one continuous wire, capital letters clockwise; (3) wind the first layer tight against the tooth from the slot bottom outwards at about 10–15 % of the wire’s breaking load, filling only half of each slot; (4) count turns exactly; (5) terminate, form the star point and lace the end turns; (6) test before impregnation: phase resistances within $\pm 2\%$, insulation $> 100 \text{ M}\Omega$ at 500 V, inductances within $\pm 5\%$, and, once the rotor is fitted, three equal back-EMF sine waves; (7) impregnate with class-H varnish (pre-heat 100°C , cure about 2 h at 150°C) and re-test. Hand winding takes 3–5 h per JXA-120 stator, a manual needle winder 20–40 min and a CNC multi-spindle winder 1–3 min.

B. Other parts and assembly

Laminations are fibre-laser or wire-EDM cut M270-35A (bonded, powder-coated); magnets are bought pre-magnetised N42SH arc segments bonded with high-temperature epoxy using a non-magnetic spacer comb;

the rotor can is turned EN3/EN8 steel; gears are hobbled EN353 or 20MnCr5, carburised to 58–62 HRC, DIN 7; the ring gear is wire-EDM cut; output bearings are crossed-roller or paired thin-section units; the driver is an STM32G4 board with a three-phase gate driver, 80–100 V MOSFETs and a CAN transceiver, keeping RobStride’s MIT-mode protocol. The rotor must be inserted with a guide sleeve because magnetic pull exceeds several hundred newtons; the air gap is only 0.55 mm, so the hub must be concentric with the rotor bearings to $\leq 0.02 \text{ mm}$.

C. Casting and moulding

Molten aluminium (poured at $700\text{--}750^\circ\text{C}$) cannot be cast in silicone moulds, whose best grades survive about 300°C , and must never be poured into cement or damp media, because water flashes to steam at about 1,600 times its volume [28]. Silicone moulds suit urethane resins or low-melting alloys only; plastic injection moulding (moulds INR 1.5–15 lakh) suits covers but cannot hold $\pm 0.01 \text{ mm}$ bearing seats. For aluminium housings we compare five routes with

$$C_{\text{part}}(n) = \frac{C_{\text{tool}}}{n} + c_{\text{blank}} + t_m r(n), \quad (8)$$

where t_m is the finish-machining time (25–60 % lower for a near-net blank) and $r(n)$ the shop rate, falling from INR 2,500/h for tiny lots to INR 400/h at volume. Table VII and Fig. 4 show that billet machining wins below about 10 pieces, sand casting from a 3D-printed pattern from about 20 pieces, and high-pressure die casting only near 10,000 pieces. The saving is modest (about INR 400–500 per housing, 1.5–2 % of a batch actuator) because the bearing seats and two set-ups dominate machining either way. Sintered or metal-injection-moulded planets (INR 20–150 each after tooling) are the larger lever at volume, while the highly loaded sun should remain hobbled and case-hardened.

D. Test protocol

Each design iteration is verified by: winding resistance and inductance balance; insulation and hipot; back-EMF and harmonic content; cogging torque ($< 1\%$ of rated after compensation); K_t and peak torque with a lever arm and load cell; torque–speed on a hysteresis brake at 48 V ($\pm 10\%$ of Fig. 1); a 1 h thermal soak at continuous torque ($< 120^\circ\text{C}$); backlash ($< 15 \text{ arcmin}$); and a life test replaying JX1 gait torque traces for $\geq 200 \text{ h}$ at 50–80 % of rated torque. The longest published life test of a DIY QDD reducer is 57 h [26].

VII. COST ANALYSIS

A. Method

Unit cost is the sum of parts, labour hr_L and a test/scrap allowance; the loaded cost adds equipment written off over three years and overhead shared over the yearly volume N :

$$C_{\text{loaded}} = \sum_i c_i + hr_L + a + \frac{C_{\text{NRE}}/3 + C_{\text{OH}}}{N}. \quad (9)$$

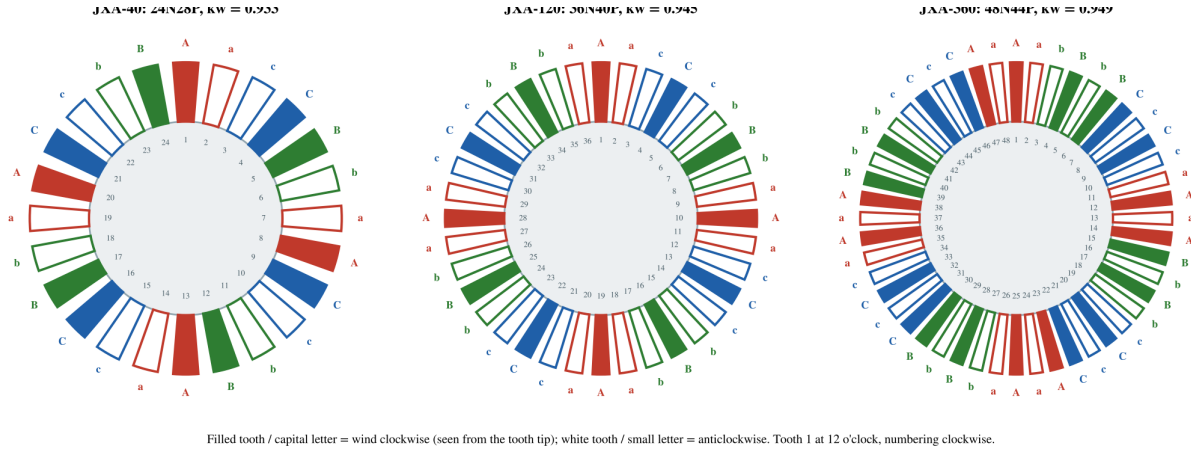


Fig. 3. Winding layouts from the star of slots, viewed from the tooth tips: each tooth carries one coil; capital letter = clockwise, lower case = anticlockwise; tooth 1 at 12 o'clock, numbered clockwise.

TABLE VI
WINDING DATA (CLASS-200 DUAL-COAT WIRE; STAR CONNECTION)

Class	Teeth	Turns/tooth	Strands $\times$ wire [mm]	Paths	Wire [m] / Cu [g]	R_{ph} 20 °C [m Ω]	Pattern from tooth 1 (one repeating unit)
JXA-40	24	7	2 $\times$ 0.63	1	16 / 45	69	A a c C B b a A C c b B ($\times 2$)
JXA-120	36	19	1 $\times$ 0.60	4	41 / 104	48	A a c C c b B b a ($\times 4$)
JXA-360	48	4	3 $\times$ 0.85	2	45 / 230	12	A a b B b B C c C c a A a A B b B b c C c C A a ($\times 2$)

TABLE VII
COST PER FINISHED JXA-120 HOUSING [INR] (CHEAPEST IN BOLD)

Route / lot size	2	25	200	1k	10k
CNC from billet	5,960	2,004	1,242	1,034	1,029
Sand cast + machine	8,426	1,585	732	548	536
Gravity die + machine	65,895	6,238	1,399	767	652
HPDC + machine	205,350	17,072	2,516	792	425
DMLS AISi10Mg + machine	12,620	10,762	10,436	10,350	10,348

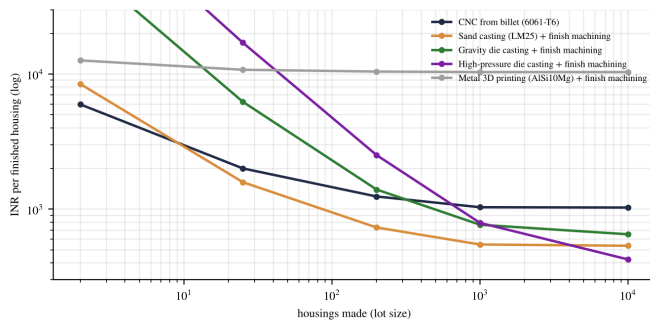


Fig. 4. Finished-housing cost versus lot size for five process routes.

Three tiers are modelled: prototype (2 units, retail parts, engineer at INR 500/h), batch (25/yr, job-work parts, technician at INR 300/h) and small factory (1,000/yr, stamped laminations, bulk magnets and cores, operator at INR 220/h). Prices are converted at INR 95.96/US\$ and INR 13.52/CNY, with imports carrying duty (32.3%) and freight (7%). The Chinese manufacturer's cost (COGS)

is estimated from list prices less a 30–45% gross margin and cross-checked with broker teardowns of the Unitree G1 [22], [23].

B. Results

Table VIII and Fig. 5 give the per-unit results, and Table IX the JXA-120 line items (Fig. 6). A prototype JXA-120 costs INR 77,353, a batch unit INR 30,292 and a factory unit INR 11,493, against INR 34,634 (US-dollar list, landed) or INR 24,091 (China distributor, landed) for RS04, and an estimated INR 11,357 for RobStride's own cost. Including equipment and overhead, the loaded JXA-120 costs INR 95,241, INR 53,279 and INR 15,167 in the three tiers.

C. Equipment

The prototype kit (winding jig, LCR and milliohm meters, insulation tester, supply, oven, oscilloscope, a hysteresis-brake dynamometer with torque sensor) costs INR 3.2 lakh; batch add-ons (semi-automatic needle winder, surge tester, second dynamometer, fixtures, inspection) INR 12.5 lakh; and a small factory (CNC winders, stamping dies, gear tooling, end-of-line testers, balancing, varnish line) INR 1.35 crore (Appendix A).

D. Scenarios and break-even

For JX1's 13 leg and waist actuators, buying costs INR 399,310 (USD route) or INR 255,040 (China distributor). Building 15 in-house units (one spare per class)

TABLE VIII
DIRECT COST PER ACTUATOR VERSUS BUYING [INR] (ESTIMATED)

Class	Make in India			Buy (landed in India)		China maker's COGS (est.)
	Prototype	Batch 25/yr	1,000/yr	USD list	China distributor	
JXA-40	52,845	19,777	7,479	28,522	17,059	6,354–13,520
JXA-120	77,353	30,292	11,493	34,634	24,091	8,923–20,280
JXA-360	146,564	61,822	23,322	65,194	–	20,280–40,560

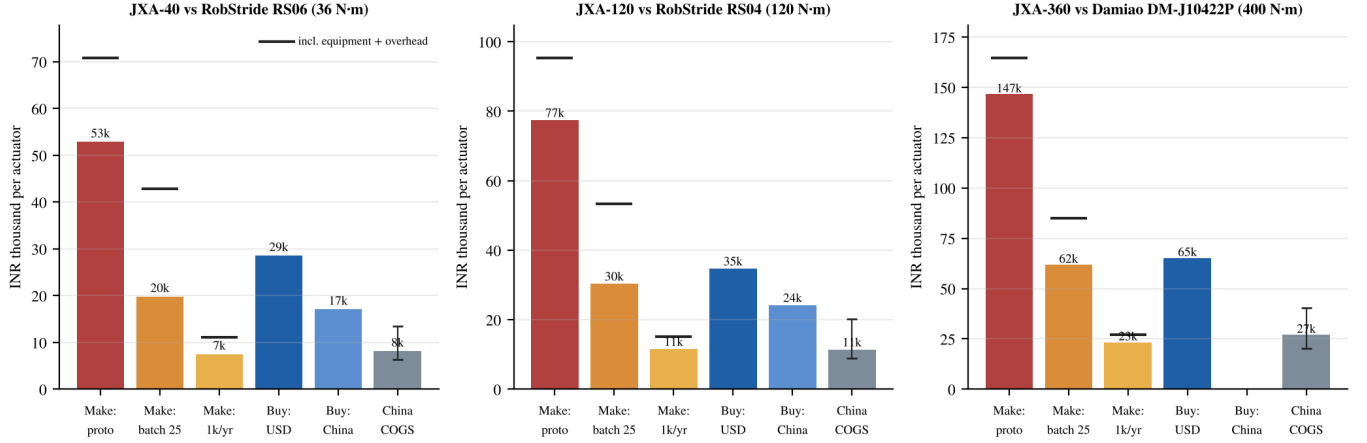
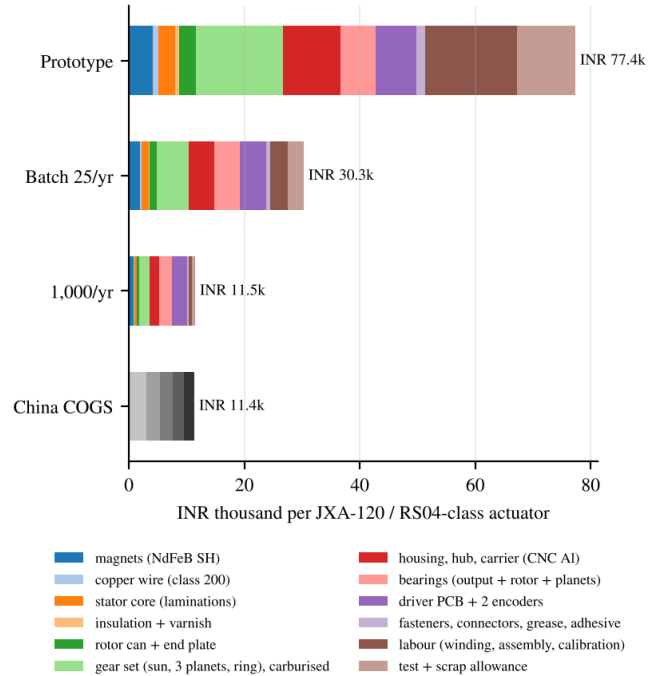


Fig. 5. Per-unit cost: making (prototype, batch, 1,000/yr) versus buying versus the estimated cost to a Chinese manufacturer (with range). Horizontal bars include equipment (three-year write-off) and overhead.

TABLE IX
JXA-120 COST LINE ITEMS [INR]

Item	Prototype	Batch	1k/yr
magnets (NdFeB SH)	4,128	1,876	751
copper wire (class 200)	935	312	265
stator core (laminations)	3,000	1,200	300
insulation + varnish	600	250	80
rotor can + end plate	3,000	1,200	400
gear set (sun, 3 planets, ring), carburised	15,000	5,500	1,800
housing, hub, carrier (CNC Al)	10,000	4,500	1,600
bearings (output + rotor + planets)	6,100	4,400	2,250
driver PCB + 2 encoders	7,000	4,500	2,600
fasteners, connectors, grease, adhesive	1,500	800	350
labour (winding, assembly, calibration)	16,000	3,000	550
test + scrap allowance	10,089	2,754	547
Total	77,353	30,292	11,493



costs INR 370,258 in parts and labour, plus INR 520,789 for two prototype rounds and INR 322,000 of equipment, INR 1,213,048 in total; each further set costs INR 320,190. For a full-size humanoid with 4×360 , 14×120 and 10×40 N·m joints, buying costs INR 1,030,884 (or INR 768,647); building costs INR 2,197,637 at prototype prices, INR 869,142 (loaded INR 1,512,769) at batch prices and INR 328,974 (loaded INR 431,864) at factory prices, against an estimated INR 348,275 for a Chinese manufacturer.

The break-even volume against the landed China-

Fig. 6. Where the money goes in a JXA-120 at the three tiers, and the estimated split of a Chinese manufacturer's cost.

distributor price P is

$$N^* = \frac{C_{NRE}/3 + C_{OH}}{P - C_{unit}}. \quad (10)$$

A. Prototype kit (INR 3.2 lakh)	
winding jig + turn counter	5,000
LCR meter (TH2830 class)	45,000
insulation (hipot) tester	15,000
4-wire milliohm meter	10,000
bench supply 60 V / 15 A	18,000
curing oven (used)	15,000
oscilloscope	35,000
dyno bench: hysteresis brake + DYN-200 torque sensor + frame	140,000
USB-CAN adapters	6,000
printed assembly fixtures, rotor guide sleeves	8,000
hand tools, consumables	25,000
B. Batch add-ons (INR 12.5 lakh)	
semi-automatic needle winder (US\$5-8k + duty)	700,000
surge (impulse) tester	100,000
second dyno + thermal chamber	150,000
assembly + magnet-bonding fixtures	150,000
simple rotor balancer	50,000
inspection: bore gauges, height gauge, gear-roll tester	100,000
C. Small-factory add-ons (INR 1.35 crore)	
CNC needle winders x2	4,000,000
stator stamping dies, 3 classes (no quote found)	2,400,000
gear tooling (hobs, PM dies for sun/planets)	1,500,000
end-of-line test stations x2	2,000,000
dynamic balancing machine	600,000
varnish/cure line	500,000
ESD assembly line + fixtures	1,500,000
quality lab	1,000,000

At batch scale the JXA-120 costs INR 6,200 *more* than buying before any equipment, so no break-even exists. At factory scale the margin is INR 12,599 per unit against INR 1.10 crore of yearly fixed cost, giving $N^* \approx 875$ units per year. Even then an Indian line does not reach RobStride’s estimated own cost and must compete on duty, service, lead time and supply security. The 360 N·m class is the exception: a batch JXA-360 (INR 61,822) already matches the landed price of the nearest import (INR 65,194).

VIII. DISCUSSION

Model accuracy. The analytic model omits finite-element saturation, harmonic and eddy losses in magnets, and detailed thermal paths; the 3–20% agreement with RS04 is encouraging but a single comparison. *Supply risk.* High-temperature (SH) NdFeB grades containing Dy/Tb need Chinese export licences under the April 2025 rules, and broader rules are suspended only until 10 November 2026; India’s INR 7,280 crore magnet scheme has no producing plant yet [25]. Magnets cost only about INR 751 per JXA-120 at bulk prices, so the risk is availability, not cost. *Economics.* Chinese cost advantage comes from volume (RobStride shipped over 50,000 joints by September 2025 and targets 300,000 in 2026 [24]) and a dense supply chain, not from wages, which are higher in China than in India; direct labour is only 8–14% of Unitree’s cost [23].

IX. CONCLUSION

A RobStride-class QDD actuator can be designed from first principles and built in India with job-shop processes,

bought stator cores and hand winding; the analytic JXA-120 reproduces the RS04 datasheet within 3–20%. A INR 3–4 lakh test kit suffices for prototypes. In-house units are not cheaper below about 875 units per year, so JX1 should continue to use imported actuators while the JXA family is developed in parallel, starting with the 360 N·m class, where no low-cost import exists, and with supply security as the goal. Future work is the construction and bench testing of JXA-120 prototypes, finite-element refinement of the motor, and a life test of the planetary stage.

APPENDIX A EQUIPMENT LISTS [INR]

APPENDIX B REPRODUCIBILITY

All results are generated by `actuators/inhouse/qdd_design.py` (design, figures), `cost_model.py` (costs) and `make_paper.py` (this paper). Source data and every price, with URL, date and evidence label, are in the research logs [26]–[30].

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